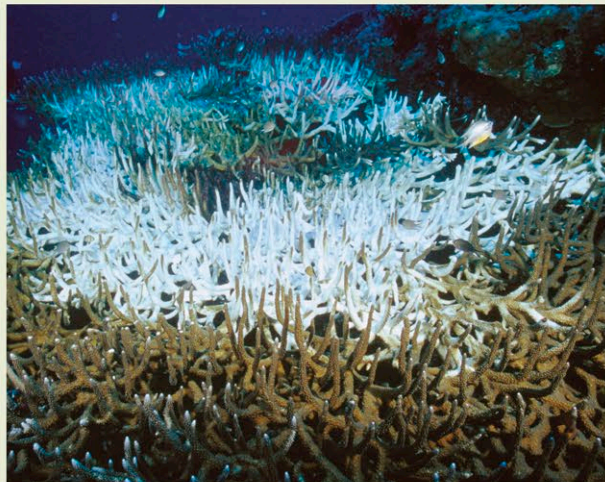


Coasts and coral reefs

In the natural world, the boundaries between different environments are often rich habitats for wildlife. The world's coasts are the ultimate example of this meeting of habitats because they bring together animals that live on the land and those that live in the sea. Coastal wildlife varies according to local conditions but, on rocky coasts as well as sandy ones, shore animals are associated with clear-cut zones that are usually determined by the tides. Coral reefs are a special kind of coastal habitat. Famed for their spectacular shape and color, they can grow to vast proportions and are unrivaled in the immense variety of aquatic life that they support. Reef-building corals require specific conditions, the most important being warmth and bright sunlight all year round. As a result, coral reefs are largely restricted to the tropics.

MARINE POLLUTION

With their abundant wildlife, coasts and coral reefs are easily damaged by pollution. Among the most visible pollutants are crude oil, which can be fatal to surface-dwelling animals, and nonbiodegradable plastics, which litter shallows and shores. Pollution can also affect the water invisibly, when contaminants such as industrial effluents are washed into the sea. On a global scale, marine life faces larger problems: the gradual warming and acidification of seawater, caused by increasing levels of carbon dioxide in the earth's atmosphere. In coral reefs, periodic warming sometimes triggers bleaching, which occurs when corals expel the symbiotic algae that provide them with much of their food. Corals often recover from bleaching, but acidification is a more long-term and pervasive change. It makes it harder for marine animals to extract calcium carbonate—the key mineral that they use for building skeletons and shells.



BLEACHED CORALS

Reaching upward like skeletal fingers, bleached *Acropora* coral contrasts with the living coral around it. Bleaching is not necessarily fatal to corals, but it is a sign of a reef under stress—something that is becoming more common as global sea temperatures rise.

Coasts

Some habitats look much the same from one year to the next, but the seashore is always changing. Waves pound away at rocks, undermining them and breaking them up, while coastal currents reshape the shore in a less dramatic way by moving pebbles and sand. Superimposed on this is the rhythmic movement of the tides—a twice-daily cycle that has a profound impact on seashore wildlife.

Tides vary enormously in different parts of the world. Around islands in mid-ocean, the total rise and fall is often less than 12 in (30 cm), while in deep bays and inlets on continental coasts, it can be over 33 ft (10 m). Whatever its height range, the tide divides the shore into 3 different zones, each with its own distinctive animal life.

The highest of these zones is the supralittoral, which is the part of the shore just beyond the reach of the highest tides. Although this zone is never actually submerged, it is affected by salt spray, which means that animals that are sensitive to salt—and also salt-intolerant plants—are rarely found in this zone. Below this is the eulittoral, a zone that is regularly covered and then exposed as the tide floods and recedes. The animals found here, such as mussels and limpets, lead a double life in that they have to be able to survive both in water and in air. The next zone, or sublittoral, is always submerged, even during the lowest tides. Most of the animals that live in this zone are fully marine, although a few leave the water to breed.

Coastal wildlife is also affected by the geology of the shore. Many animals live on rocky coasts, while others specialize in living in sand or coastal mud. Compared with these, shingle is a difficult habitat for animals, although some shoreline waders use it as a place to nest.

CLIFFS are important nesting sites for many seabirds, although some, such as terns, nest on the ground above the high-tide line.

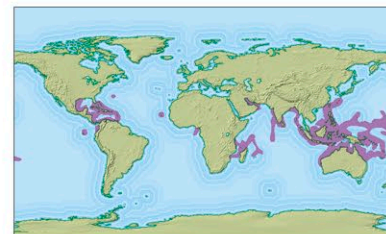
SUPRALITTORAL



Coral reefs

There are 2 types of corals: hard and soft. Coral reefs are formed by hard corals. The individual coral animals, called polyps, secrete external skeletons that persist after they die. Soft corals are found all over the world, but hard, reef-building corals grow in clear, nutrient-poor water, where their mutualistic algae can use the energy in sunlight. Deep-water coral reefs do not rely on algae. Such is the richness of the habitat created by coral reefs that a huge number of animals are able to live side by side without competing for the same food.

There are 3 main types of reefs: fringing reefs, which grow close to the shore; barrier reefs, which are separated from the coast by deep channels, sometimes over 62 miles (100 km) wide; and atolls, which are ring- or horseshoe-shaped reefs that grow around oceanic islands, often where volcanoes have subsided into the sea.



CORAL REEFS are found in parts of the world where the sea temperature never drops below about 64°F (18°C). The world's largest reefs are in the Indo-Pacific region.

Although every reef is unique, reefs share a common structure. Depth and exposure are 2 important factors that influence this because corals vary in their need for light and in their ability to withstand the force of the waves. The fastest-growing corals, which need lots of light and relatively calm water, typically form the central zone of the reef, projecting just above the surface at the lowest tides. On its inshore edge, the central zone is often backed by lagoons—large pools of open water lying over coral sand. By contrast, the seaward edge of the reef often forms a submarine cliff that drops steeply into the depths. The corals in this zone are solid and resilient because they have to withstand heavy breakers rolling in from the open sea.



LAGOON

RAYS, and many other bottom-dwelling fish, feed on animals buried in the coral sand of reef lagoons. Most are well camouflaged.

